Robust Unit Commitment with Multi-State Uncertainty: A Novel Formulation and Scalable Solution Method

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APPENDIX A
PROOF OF PROPOSITION 1

Proof: Firstly, for the sake of description convenience, three auxiliary uncertainty sets U_2', U_0, U_0' are designed as below.

$$U_2' = \Big\{ \boldsymbol{w} \in \mathbb{R}^{N_w \times N_t}: \tag{A.1a}$$

$$w_{r,t} = \sum\nolimits_{i=0}^{N_s} \omega_{r,t}^{i} y_{r,t}^{i}, \forall r, \forall t \tag{A.1b}$$

$$\sum\nolimits_{i=0}^{N_s} y_{r,t}^{i} = 1, \forall r, \forall t \tag{A.1c}$$

$$\sum\nolimits_{t=1}^{N_t} \sum\nolimits_{i=0}^{N_s} \tau_{r,t}^{i} y_{r,t}^{i} \le \Gamma_r, \forall r \tag{A.1d}$$

$$y_{r,t}^{i}, y_{r,t}^{0} \in \{0,1\}, \forall r, \forall t, \forall i \Big\} \tag{A.1e}$$

$$U_0 = \{ \boldsymbol{w} \in \mathbb{R}^{N_w \times N_t}: \tag{A.2a}$$

$$w_{r,t} = \bar{\omega}_{r,t} + \delta_{r,t}^{+} y_{r,t}^{+} + \delta_{r,t}^{-} y_{r,t}^{-}, \forall r, \forall t \tag{A.2b}$$

$$y_{r,t}^{+} + y_{r,t}^{-} \le 1, \forall r, \forall t \tag{A.2c}$$

$$\sum\nolimits_{t=1}^{N_t} \left(y_{r,t}^{+} + y_{r,t}^{-} \right) = \Gamma_r, \forall r \tag{A.2d}$$

$$y_{r,t}^{+}, y_{r,t}^{-} \in \{0,1\}, \forall r, \forall t \Big\} \tag{A.2e}$$

$$U_0' = \Big\{ \boldsymbol{w} \in \mathbb{R}^{N_w \times N_t}: \tag{A.3a}$$

$$w_{r,t} = \bar{\omega}_{r,t} + \delta_{r,t}^{+} z_{r,t}^{+} + \delta_{r,t}^{-} z_{r,t}^{-}, \forall r, \forall t \tag{A.3b}$$

$$z_{r,t}^{+} + z_{r,t}^{-} \le 1, \forall r, \forall t \tag{A.3c}$$

$$\sum\nolimits_{t=1}^{N_t} \left(z_{r,t}^{+} + z_{r,t}^{-} \right) \le \Gamma_r, \forall r \tag{A.3d}$$

$$0 \le z_{r,t}^{+}, z_{r,t}^{-} \le 1, \forall r, \forall t \Big\} \tag{A.3e}$$

where the same symbols in U_2' , U_0 , U_0' as those in U_1 described by (2a)-(2e) and U_2 described by (3a)-(3f) have the same meaning; $y_{r,t}^{0}$ denotes if the uncertain WPO takes the 0-th state values ($y_{r,t}^{0} = 1$); $\omega_{r,t}^{0}$ denotes the 0-th state values, and equals to the forecasted value $\bar{\omega}_{r,t}$; $\tau_{r,t}^{0}$ denotes the relative deviation of the 0-th state values, and equals to 0 according to (4); $z_{r,t}^{+}$ and $z_{r,t}^{-}$ are variables and denote the positive and negative relative deviation of $w_{r,t}$ from the forecasted value $\bar{\omega}_{r,t}$,respectively.

Secondly, taking advantage of the construction characteristics and differences among U_2', U_0, U_0', U_1 and U_2, the deduction process is composed of six parts as below.

1) $U_2 \subseteq U_2'$. To prove that U_2 is a subset of U_2', we need to show that for any element in U_2, it also belongs to U_2'. Let $\tilde{\boldsymbol{w}}$ be an arbitrary element in U_2. According to the definition of U_2, this implies that there exist binary variables $\tilde{\boldsymbol{y}} = (\tilde{y}_{r,t}^{i})$, $i = 1, \cdots, N_s$, such that $\tilde{\boldsymbol{w}} = \sum\nolimits_{i=1}^{N_s} \omega_{r,t}^{i} \tilde{y}_{r,t}^{i}$ satisfies the constraints (3a) - (3f). To show that $\tilde{\boldsymbol{w}}$ also belongs to U_2', we need to show that there exists binary variable $\boldsymbol{y} = (y_{r,t}^{0}, y_{r,t}^{i})$, $i = 1, \cdots, N_s$ such that $\tilde{\boldsymbol{w}} = \sum\nolimits_{i=0}^{N_s} \omega_{r,t}^{i} y_{r,t}^{i}$ satisfies the constraints (A.1a)-(A.1e). By comparing the constraints (3a) - (3f) and (A.1a)-(A.1e), we observe that when $y_{r,t}^{0} = 0$ and $y_{r,t}^{i} = \tilde{y}_{r,t}^{i}$, $i = 1, \cdots, N_s$, there always exists binary variable $\boldsymbol{y} = (0, \tilde{y}_{r,t}^{i})$, $i = 1, \cdots, N_s$, such that $\tilde{\boldsymbol{w}} = \sum\nolimits_{i=0}^{N_s} \omega_{r,t}^{i} y_{r,t}^{i} = \sum\nolimits_{i=1}^{N_s} \omega_{r,t}^{i} \tilde{y}_{r,t}^{i}$ satisfies the constraints (A.1a)-(A.1e). Thus, it can be concluded that for any element $\tilde{\boldsymbol{w}}$ in U_2, it also belongs to U_2'. Equivalently, $U_2 \subseteq U_2'$.

2) $U_0 \subseteq U_1 \subseteq U_2' \subseteq U_0'$. This formula can be derived by the same deduction processes as that for $U_2 \subseteq U_2'$.

3) $\mathrm{vert}(U_0') = U_0$, which means that U_0 is the set of extreme points of U_0'. The sets U_0 and U_0' differ in two main aspects. First, the variables $y_{r,t}^{+}$ and $y_{r,t}^{-}$ in U_0 are restricted to binary values $\{0,1\}$, while the variables $z_{r,t}^{+}$ and $z_{r,t}^{-}$ in U_0' are continuous in the interval $[0,1]$. This difference in variables force the elements of U_0 to lie at the "corners" of the feasible region defined by U_0'. Second, the constraints (A.2d) in U_0 are equality constraints, while the corresponding constraints (A.3d) in U_0' are inequality constraints. The equality constraint (A.2d) restricts the regions defined by U_0 to be a subset of the region defined by U_0' . The binary constraints force the elements of U_0 to lie at the "corners" of U_0''s feasible region, while the equality constraint (A.2d) reinforces this property by tightly restricting the feasible space. Thus, $\mathrm{vert}(U_0') = U_0$.

4) $\mathrm{conv}(U_0) = \mathrm{conv}(U_0')$, which means that the convex hull of U_0 equals to the convex hulls of U_0'. According to the property of convex hulls, the convex hull of the extreme points of a set is equal to the convex hull of the set itself. Thus, it follows that

$\text{conv}(\text{vert}(U_0')) = \text{conv}(U_0')$. Combining with the earlier result $\text{vert}(U_0') = U_0$ derived at the third part, it follows that $\text{conv}(\text{vert}(U_0')) = \text{conv}(U_0)$. Thus, $\text{conv}(U_0)=\text{conv}(U_0')$.

5) $\text{conv}(U_1) = \text{conv}(U_2')$. Combining with the earlier result $U_0 \subseteq U_1 \subseteq U_2' \subseteq U_0'$ derived at the second part, it follows that $\text{conv}(U_0) \subseteq \text{conv}(U_1) \subseteq \text{conv}(U_2') \subseteq \text{conv}(U_0')$. Moreover, combining with the earlier result $\text{conv}(U_0)=\text{conv}(U_0')$ derived at the fourth part, it follows that $\text{conv}(U_0) = \text{conv}(U_1) = \text{conv}(U_2') = \text{conv}(U_0')$. Thus, $\text{conv}(U_1) = \text{conv}(U_2')$.

6) $\text{conv}(U_2) \subseteq \text{conv}(U_2')$. Combining with the earlier result $U_2 \subseteq U_2'$ derived at the first part, it follows that $\text{conv}(U_2) \subseteq \text{conv}(U_2')$.

7) Combining with the earlier results $\text{conv}(U_2) \subseteq \text{conv}(U_2')$ derived at the sixth and $\text{conv}(U_1) = \text{conv}(U_2')$ derived at the fifth part, it follows $\text{conv}(U_2) \subseteq \text{conv}(U_1)$. Thus, Proposition 1 is proven.

□

APPENDIX B

PROOF OF PROPOSITION 2

Proof: In step 3, if the termination criterion is not satisfied, then the algorithm proceeds to either the exploration or the exploitation step. Therefore, to prove the finite convergence of the algorithms, it suffices to show that: The algorithm can proceed to the exploration step and the exploitation step only a finite number of times. The deduction process is composed of two parts as below.

1) To prove that the exploration step can only be executed a finite number of times, it is indeed equivalent to showing that for any U which contains a finite number of elements or extreme points (denoted by N), the exploration step can be only executed a finite number of times before the algorithm either meets the termination criteria $(\bar{U} - L_{MP}^{\ell})/\bar{U} < \varepsilon$ or turns to the exploitation step. To proceed with a proof by contradiction, assume that the exploration step is executed infinitely many times. This implies that the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$, $\boldsymbol{w}^j \notin \mathcal{S}$ and $(\bar{U} - \bar{L})/\bar{U} \geq \tilde{\varepsilon}$ are satisfied after the exploration step is executed infinitely many times. However, since that the U only contains N elements or extreme points, and the criteria $\boldsymbol{w}^j \notin \mathcal{S}$ ensures that each execution of the exploration step adds a non-repeated elements or extreme points $\boldsymbol{w}^j$ from U to $\mathcal{S}$, the exploration step can only be executed at most N times. This directly contradicts the assumption of infinite executions. Therefore, the exploration step can only be executed a finite number of times.

2) To prove that the exploitation step can also only be executed a finite number of times, it is indeed equivalent to showing that for any fixed $\tilde{\mathcal{S}} \in U$, the exploitation step can be only executed a finite number of times before the algorithm either meets the termination criteria $(\bar{U} - L_{MP}^{\ell})/\bar{U} < \varepsilon$ or turns to the exploration step. To proceed with a proof by contradiction, assume that the exploitation step under $\tilde{\mathcal{S}}$ is executed infinitely many times. This implies that the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$ and at least one of the conditions $\boldsymbol{w}^j \in \mathcal{S}$ or $(\bar{U} - \bar{L})/\bar{U} < \tilde{\varepsilon}$ are satisfied at iteration $j = \ell$ after the exploitation step under $\tilde{\mathcal{S}}$ is executed infinitely many times. This assumption leads us to consider two separate cases, which will be analyzed in detail as below to derive a contradiction.

In the first case, assume that the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$ and $\boldsymbol{w}^j \in \mathcal{S}$ are satisfied at iteration $j = \ell$ after the exploitation step under $\tilde{\mathcal{S}}$ is executed infinitely many times. As the exploitation steps are executed, the relative optimality gap tolerance ε_{MP} and ε_{SP} of the master and subproblems gradually decreases and approaches 0. When the master and subproblems are solved exactly, the solution $\boldsymbol{w}^j$ obtained from the exact solution of the subproblem belongs to $\tilde{\mathcal{S}}$ if and only if the optimality gap between the master problem and the subproblem is 0, i.e., $\bar{U} = L_{MP}^{\ell}$. This implies that the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$ and $\boldsymbol{w}^j \in \mathcal{S}$ cannot both hold simultaneously, contradicting the initial assumption.

In the second case, assume that the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$ and $(\bar{U} - \bar{L})/\bar{U} < \tilde{\varepsilon}$ are satisfied at iteration $j = \ell$ after the exploitation step under $\tilde{\mathcal{S}}$ is executed infinitely many times. In the II-C&CG method, we adopt U_{SP}^{j} to update $\bar{U}$ in step 2.2. This approach ensures that, regardless of the magnitude of ε_{SP}, the updated $\bar{U}$ always serves as valid upper bounds for the optimal value υ^* of the problem (1). This property allows us to conclude, similarly to the I-C&CG method, that if $\tilde{\varepsilon} < \varepsilon/(1+\varepsilon)$, the conditions $(\bar{U} - L_{MP}^{\ell})/\bar{U} \geq \varepsilon$ and $(\bar{U} - \bar{L})/\bar{U} < \tilde{\varepsilon}$ cannot both hold simultaneously, contradicting the initial assumption.

In both cases, the assumptions that the exploitation step is executed infinitely many times lead to contradictions. Therefore, the exploitation step can also only be executed a finite number of times.

In summary, the algorithm can proceed to the exploration step and the exploitation step only a finite number of times. This demonstrates that the computation of the II-C&CG method terminates in a finite number of iterations. Thus, Proposition 2 is proven.

□

APPENDIX C

SUPPLEMENTARY COMPARISON RESULTS

We conduct supplementary experiments to compare the effectiveness of U_1, U_2, U_3, U_4, U_5 in reducing conservatism across the 6-bus and 118-bus systems, as well as with various settings of uncertainty set parameters, following the comparative methodology outlined in Section V-B.

In addition to the average day-ahead scheduling cost and the average intraday dispatch cost, we also supplement with another indicator, the number of days that the daily day-ahead scheduling schemes fail to pass safety verification (denoted as

n), as three indicators to assess the effectiveness of U_1, U_2, U_3, U_4, U_5. We say that a day-ahead scheduling scheme does not pass the safety verification, if the load shedding penalty cost computed from the dispatch program is larger than 0. Table II, III, IV, V, VI shows the above three indicators calculated under different parameters of U_1, U_2, U_3, U_4, U_5 respectively.

TABLE II

THREE INDICATORS CALCULATED FOR U_1

Bus	$\mathrm{Pr_I}$	Γ	c_1^1 ($)	c_2^1 ($)	n
6	0.95	6	172,438	129,227	0
		12	201,690	133,052	0
		18	224,734	133,409	0
		24	239,891	134,144	0
118		12	1,369,757	1,333,735	0
		24	1,409,315	1,336,933	0

TABLE III

THREE INDICATORS CALCULATED FOR U_2

Bus	N_s	$\mathrm{Pr_I}$	M	$\mathrm{Pr_T}$	Γ	c_1^2 ($)	c_2^2 ($)	n
6	9	0.95	20	0.9	6	164,234	122,423	0
					12	186,636	127,008	0
					18	211837	127,469	0
					24	224,651	128,807	0
				0.95	6	166,810	125,479	0
					12	192,528	129,935	0
					18	217,101	130,134	0
					24	233,007	131,319	0
				1	6	168,945	128,465	0
					12	194,897	132,772	0
					18	220,689	132,475	0
					24	238,973	130,109	0
118				0.9	12	1,351,435	1,446,563	1
					24	1,408,466	1,335,365	0
				0.95	12	1,359,268	1,329,049	0
					24	1,408,526	1,335,819	0
				1	12	1,366,584	1,333,185	0
					24	1,409,101	1,336,678	0

TABLE IV

THREE INDICATORS CALCULATED FOR U_3

Bus	α_1	Λ	Γ	c_1^3 ($)	c_2^3 ($)	n
6	0.3	1	6	75,932	2,468,000	5
			12	97,202	1,397,854	5
			18	105,419	1,369,054	5
			24	111,292	1,389,548	5
		5	6	86,370	1,404,527	5
			12	99,950	1,363,314	5
			18	107,463	1,353,739	5
			24	112,248	1,359,221	5
	1	1	6	165,238	1,80,6170	5
			12	270,804	155,534	0
			18	307,500	156,598	0
			24	332,184	157,132	0
		5	6	228,531	154,564	0
			12	277,509	154,469	0
			18	310,605	154,932	0
			24	331,940	156,318	0
118	0.3	1	12	1,325,446	8,418,346	1
			24	1,336,473	8,425,195	1
		5	12	1,325,446	8,418,346	1
			24	1,321,293	8,719,817	1
	1	1	12	1,548,564	1,355,843	0
			24	1,626,314	1,365,031	0
		5	12	1,555,804	1,345,337	0
			24	1,627,783	1,356,596	0

TABLE V

THREE INDICATORS CALCULATED FOR U_4

Bus	N_b	α_1	α_2	Λ	Γ	c_1^4 ($)	c_2^4 ($)	n
6	4	0.3	0.08	1	6	65,112	2,833,062	6
					12	79,799	1,993,041	6
					18	85,304	1,922,467	6
					24	89,731	1,867,714	6
				5	6	71,951	1,934,815	6
					12	82,067	1,898,995	6
					18	86,984	1,922,299	6
					24	90,653	1,925,514	6
		1	0.25	1	6	128056	2,382,343	5
					12	203916	591,514	5
					18	244867	152,125	0
					24	262731	152,556	0
				5	6	197782	154,686	0
					12	222899	154,535	0
					18	249022	154,379	0
					24	265526	154,293	0
118		0.3	0.08	1	12	1,305,930	8,382,304	1
					24	1,316,179	8,443,149	1
				5	12	1,304,202	8,401,256	1
					24	1,304,040	8,549,558	1
		1	0.25	1	12	1,332,122	1,366,817	0
					24	1,497,036	1,352,937	0
				5	12	1,441,116	1,346,558	0
					24	1,503,226	1,350,410	0

TABLE VI

THREE INDICATORS CALCULATED FOR U_5

Bus	c_1^5 ($)	c_2^5 ($)	n
6	254,840	164,220	0
118	1,413,425	1341,772	0

From Table II, III, IV, V, VI we can draw the same conclusions as in Section V-B. Due to the lack of comprehensive parameter setting strategies for U_3, U_4, the scheduling schemes based on U_3, U_4 are either unreliable or conservative. We can observe that when the prespecified parameters of U_3, U_4 are set the same to the parameters used in the numerical experiments of [5], [8] respectively, i.e. the prespecified parameters of U_3 are set as $\alpha_1 = 0.3$, $\Lambda = 5$ and the prespecified parameters of U_4 are set as $N_b = 4$, $\alpha_1 = 0.3$, $\alpha_2 = 0.08$, $\Lambda = 5$, the scheduling schemes based on U_3, U_4 lead to serve load shedding in the intraday dispatch program. Meanwhile, we can observe that when the boundary value and interval ranges of U_3, U_4 are expanded, i.e., $\alpha_1 = 1$, $\alpha_2 = 0.25$, the scheduling schemes based on U_3, U_4 don't lead to load shedding in the intraday dispatch program, which validates the robustness of the scheduling scheme based on U_3, U_4. However, the day-ahead scheduling costs and intraday cost under U_3, U_4 are much higher than thoes under U_1, U_2 at the same Γ, which suggests that the scheduling schemes based on

U_3, U_4 are over-conservative than U_1, U_2. Thus, we can find that the robustness or economics of the scheduling scheme based on U_3, U_4 can't be guaranteed due the empirical parameter setting strategies of U_3, U_4.

In contrast, the scheduling schemes based on U_1, U_2, U_5 don't lead to load shedding in the intraday dispatch program, which validates the robustness of the scheduling schemes based on U_1, U_2, U_5. The boundary value parameterized based on the quantile regression technique and the transition indicator parameterized based on the Markov chain analysis scientifically ensure the robustness of the scheduling scheme based on U_1, U_2 with confidence level $\Pr_{\mathrm{I}}$ and $\Pr_{\mathrm{T}}$. Convex hull analysis ensures the robustness of U_5. Meanwhile, the day-ahead scheduling cost and intraday dispatch cost based on U_2 is lower than that based on U_1, U_5, which demonstrates the economics of the scheduling scheme based on U_2. A more refined formulation of the WPO uncertainties and a more refined parameter setting strategy ensure the economics of U_2.

Appendix D

Verification of the Effectiveness of MSUS in MSRUCM

First, U_1, U_2, U_3, U_4, U_5 for each day of the week gained at Section V-B are utilized. Next, we solve the MSRUCM [15] based on U_1, U_2, U_3, U_4, U_5, respectively, to generate daily day-ahead scheduling schemes and costs for this week. MSRUCM based on U_1, U_2, U_3, U_4, U_5 are solved using affine policy and constraint generation method [15].

Afterwards, we substitute each day's day-ahead scheduling schemes of this week into an intraday dispatch program to generate intraday dispatch schemes and costs. The cost coefficients are consistent with those set in Section V-B. We take the average day-ahead scheduling cost of the week (denoted as d_1), and the average intraday dispatch cost of the week (denoted as d_2) and the number of days that the daily day-ahead scheduling schemes fail to pass safety verification (denoted as n) as three indicators to assess the effectiveness and advantage of U_1, U_2, U_3, U_4, U_5. d_1^i, d_2^i ($i = 1,2,3,4,5$) represent the specific cost under the corresponding U_i ($i = 1,2,3,4,5$).

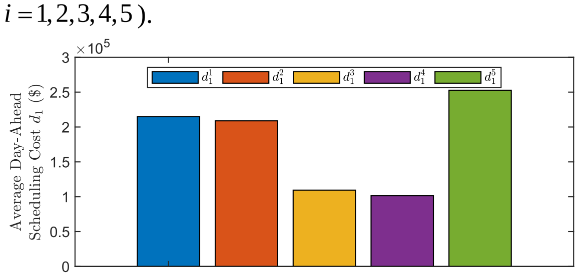

Fig. 9. Average day-ahead scheduling cost d_1^i at $\Gamma = 12$

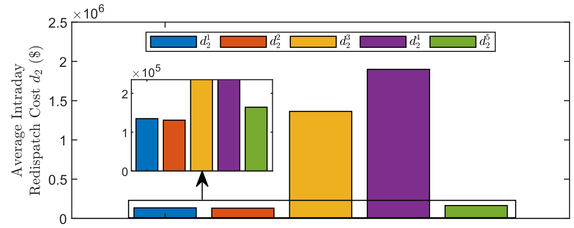

Fig. 10. Average intraday dispatch cost d_2^i at $\Gamma = 12$

From Fig. 9, 10, we can observe that the day-ahead and intra-day computational results of the MSRUCM based on U_1, U_2, U_3, U_4, U_5 show similar outcomes to those of the TSRUCM. Due to the lack of comprehensive parameter setting strategies for U_3, U_4, the scheduling schemes based on U_3, U_4 are not robust enough, leading to the occurrence of load shedding in the intra-day dispatch and expensive intra-day costs. In contrast, the scheduling schemes based on U_1, U_2, U_5 don't lead to load shedding in the intraday dispatch program, which validates the robustness of the scheduling schemes based on U_1, U_2, U_5. Meanwhile, the day-ahead scheduling cost and intraday dispatch cost based on U_2 is lower than that based on U_1, U_5, which demonstrates the economics of the scheduling schemes based on U_2. Therefore, MSUS is also effective and advanced in reducing the conservatism of the scheduling schemes generated from the MSRUCM.

Appendix E

Verification of the Effectiveness of TSDRUCM

We construct a Wasserstein-matric based AS [21]. First, we generated the original support scenario set for AS by sampling based on the probability error matrix obtained from historical data statistics. Subsequently, we used a Wasserstein distance-based scenario reduction method to reduce the original support scenario set, creating a typical support scenario set. Finally, we constructed AS based on this typical support scenario set. Fig. 11 shows the forecasted WPO scenario for Jan. 1, 2006, along with 100 WPO scenarios from the original support scenario set generated based on the prediction error probability sampling.

Next, we solve the TSDRUCM [21] based on the Wasserstein-metric based AS to generate daily day-ahead scheduling schemes and costs for the first week of January 2006 on the 6-bus system. Afterwards, we substitute each day's day-ahead scheduling schemes of this week into an intraday dispatch program to generate intraday dispatch schemes and costs. The cost coefficients are consistent with those set in Section V-B. We take the average day-ahead scheduling cost of the week (denoted as e_1), the average intraday dispatch cost of the week (denoted as e_2) and the number of days that the daily day-ahead scheduling schemes fail to pass safety verification (denoted as n) as three indicators to assess the effectiveness and advantage of TSDRUCM based on the Wasserstein-metric based AS. Table VII presents the computational results of TSDRUCM under different numbers of typical support scenario sets.

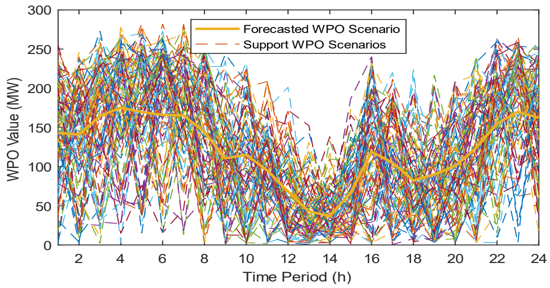

Fig. 11. The forecasted WPO scenario and support WPO scenarios

TABLE VII
THREE INDICATORS CALCULATED FOR TSDRUCM

Bus	number	e_1 (\$)	e_2 (\$)	n
6	10	126,119	4,289,485	4
	50	184,643	136,827	0
	100	189,032	138,456	0

From Table VII, we can find that when the number of scenarios in the support scenario set is small, the day-ahead scheduling scheme of TSDRUCM tends to be less robust, often leading to load shedding incidents during intraday dispatch and resulting in high intraday scheduling costs. As the number of scenarios increases, the reliability of the day-ahead scheduling scheme gradually improves, and load shedding incidents no longer occur during intraday dispatch. However, the selection of an appropriate number of scenarios remains an unresolved issue in current research, and the reliability assessment of the TSDRUCM scheduling scheme still relies on empirical judgment. When the number of wind farms is small, AS can ensure the economic efficiency and robustness of the scheduling schemes based on TSDRUCM with a relatively small number of WPO scenario samples. However, as the number of wind farms increases, the required sample size of the combinatorial WPO scenarios for multiple farms surges, making it difficult for DRUCM to guarantee the robustness of the scheduling schemes. In such cases, the scheduling schemes based on TSRUCM exhibit superior robustness and provide greater security and reliability.